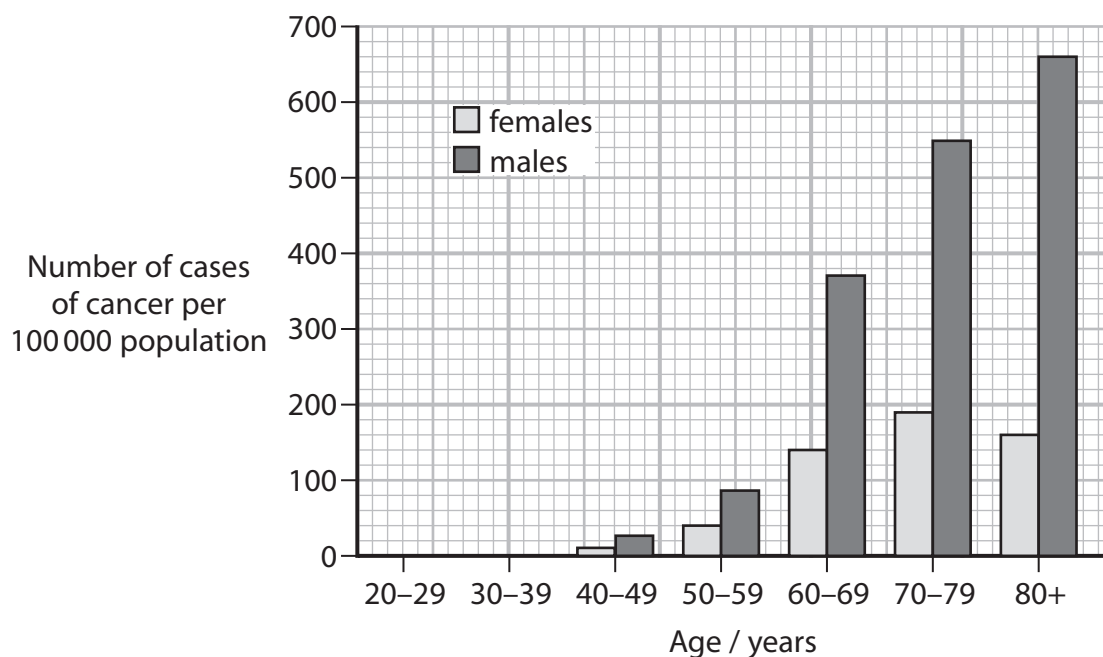


7 A change in the base sequence of a gene can give rise to cancer.

The graph shows the number of cases of one type of cancer in a human population.



(a) (i) Deduce the effect of age and sex on the number of cases of this cancer.

(3)

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- (ii) A city had a population of 2 million people. Three per cent (%) of this population were males over the age of 80.

Calculate how many of the males over the age of 80 would have this cancer.

(2)

Answer

- (b) A sample of tissue was taken from a patient with this type of cancer in order to calculate the mitotic index.

- (i) This sample was stained before being viewed using a microscope.

State why the cells in this sample were stained before being viewed using a microscope, in order to calculate the mitotic index.

(1)

- (ii) Describe how the mitotic index of the tissue sample would be determined.

(2)



(c) One treatment for this type of cancer prevents the condensing of chromosomes of cancer cells.

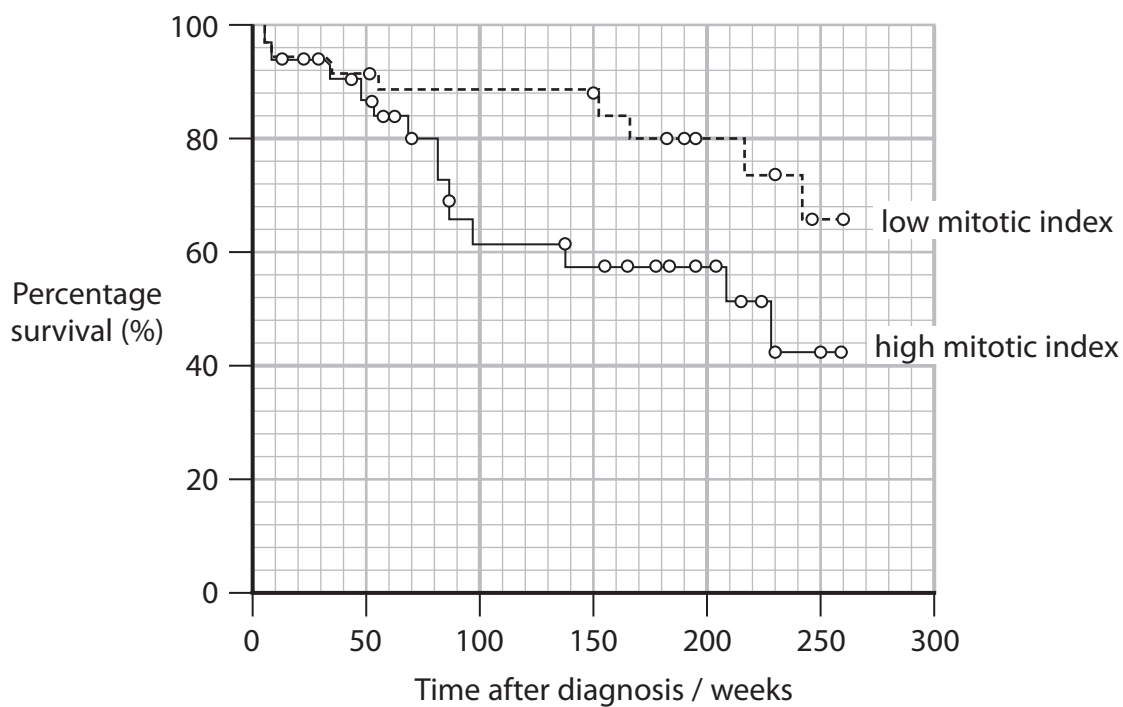
(i) In which stage of mitosis do chromosomes start to condense?

(1)

- ☐ **A** anaphase
- ☐ **B** metaphase
- ☐ **C** prophase
- ☐ **D** telophase



- (ii) The graph shows the effect of the mitotic index on the percentage survival for people with a type of cancer.



Explain how preventing the condensation of chromosomes would affect the survival of a patient with this type of cancer.

(3)

(Total for Question 7 = 12 marks)